

PERFORMANCE MANAGEMENT SYSTEM · V1.0.0

Database Guide

Entity-relationship diagrams, every table, relationships, indexes, and column-level explanations.

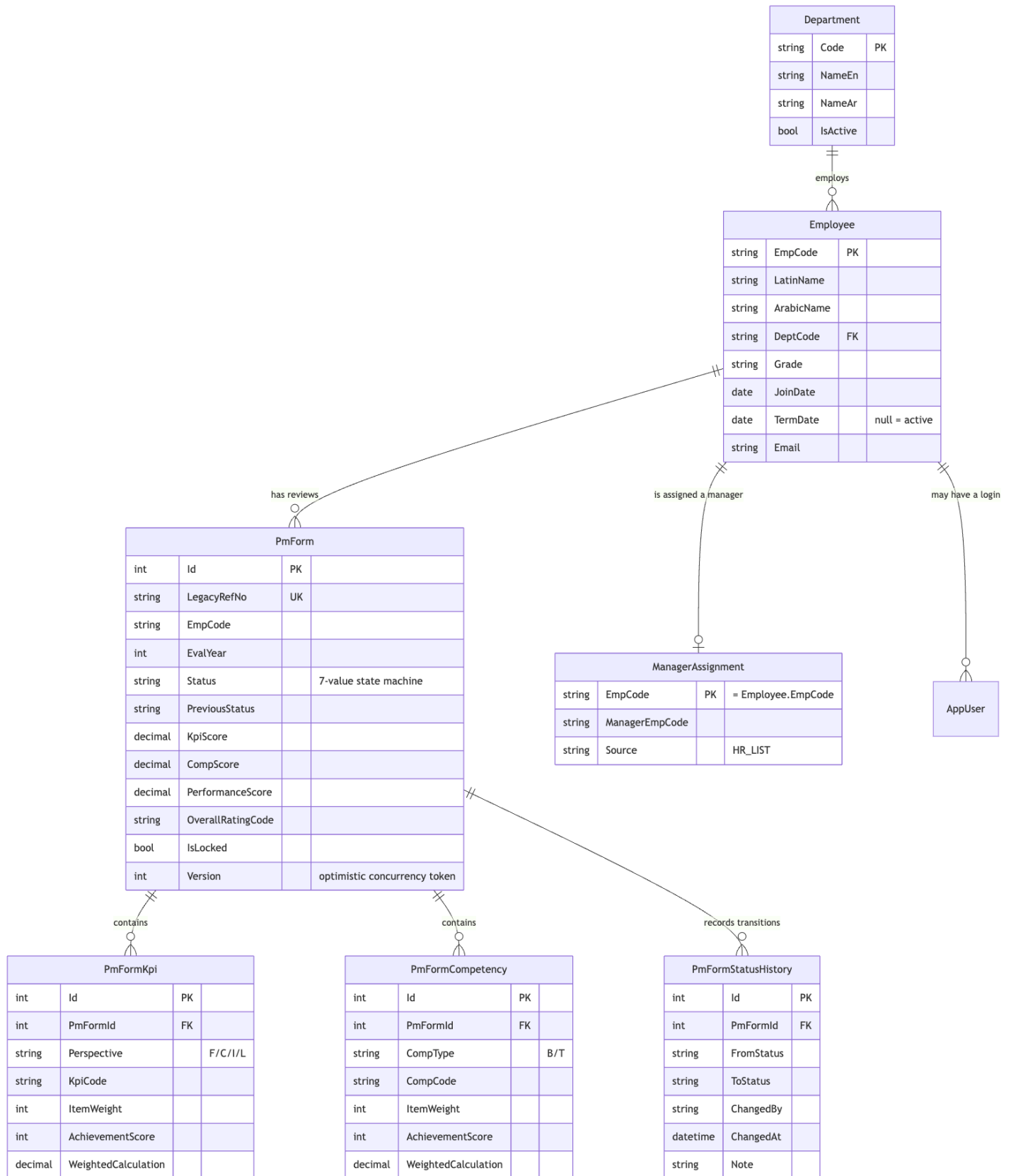
Performance Management System · Database Guide · Confidential

1.	Overview
2.	ER diagram — PM workflow core
3.	ER diagram — reference, users & system tables
4.	Table reference: workflow core
5.	Table reference: reference/master data
6.	Table reference: users, security & system
7.	Indexes
8.	Migrations

1. Overview

PostgreSQL 16, accessed exclusively through Entity Framework Core 8 (`PmDbContext`). 22 tables, applied through 11 sequential migrations (see §8). There is no direct SQL or stored-procedure logic — all business rules live in the `C#` service layer, and the schema itself is intentionally simple: one row per real-world record, snapshot fields called out explicitly where they're a deliberate design choice rather than duplication.

2. ER diagram — PM workflow core



`PmForm(EmpCode, EvalYear)` carries a unique index — the schema itself guarantees at most one review per employee per year, not just application-level convention. Name/department/grade fields on `PmForm` are point-in-time **snapshots**, not live foreign keys, by design — a completed review must always read back the way it looked when it was actually reviewed.



4. Table reference: workflow core

Employee

The PM-relevant subset of the legacy employee master. `EmpCode` (string, primary key) — the 4-digit legacy employee code, also used as the login username for non-admin accounts. `TermDate` null means active (`IsActive` is a computed property, not a stored column). `Source` distinguishes rows synthesized from the legacy import (`HDR_SNAPSHOT`) from ones entered directly in this app (`MANUAL`).

Department / Designation / Section

Simple bilingual code/name reference tables. Departments carry an `IsActive` flag (never hard-deleted — employees may reference a disabled one historically).

ManagerAssignment

One row per employee (`EmpCode` is the primary key), pointing at their HR-designated direct manager (`ManagerEmpCode`). Deliberately separate from any org-chart hierarchy — some HR-designated PM reviewers are not the employee's formal line manager.

PmForm

The authoritative record for one employee's review in one year — the legacy system's HDR (header) row. Key columns:

`Status` / `PreviousStatus` (the seven-value state machine), `KpiScore` / `CompScore` / `PerformanceScore` (computed, persisted), `IsLocked` (prevents edits mid-review at certain stages), `Version` (EF Core optimistic-concurrency token, incremented on every transition).

PmFormKpi / PmFormCompetency

Line items belonging to one `PmForm` (cascade-deleted with it). Each carries its own weight, achievement score, and computed weighted contribution — (`PmFormId`, `RecordSeq`) is unique per form.

PmFormStatusHistory

One row per state transition — the "Workflow Timeline" shown on the Workflow Administration Details page and used as the honest, literal audit of every status change (as distinct from the separate `AuditLog` table, which records *administrative* actions specifically).

5. Table reference: reference/master data

KpiMaster / CompetencyMaster

The catalog of KPIs/competencies a manager can attach to a review, each with a suggested weight range, applicable department list (`DeptCsv` , comma-separated or `*` for all), and bilingual name/description/formula text.

JobFamily

Maps a comma-separated grade list (`GradesCsv`) to a KPI/competency weight split (`KpiWeight` + `CompWeight` , must sum to 100) — this is what determines, for example, that a grade-7 "Middle Management" employee's overall score is 60% KPI-weighted and 40% competency-weighted.

RatingScale

Score-band-to-rating-label mapping (e.g. 90–94 → "Exceed Expectations"), bilingual.

EmployeeException

Data-driven, per-employee rule overrides (`RuleCode` : `PERSPECTIVE_MIN_EXEMPT` , `KPI_50_50` , `SELF_MANAGER` , `BRANCH_VIEWER`), each with an optional effective date range — lets HR grant a narrow, auditable, time-bound exception without a code change.

6. Table reference: users, security & system

AppUser / UserRole

Login accounts. `FailedLoginAttempts` / `LockedOutUntil` implement account lockout directly on the row; `PreferredCulture` persists a user's language choice across devices. `UserRole` is a simple one-to-many (a user may hold zero, one, or both of `HR_ADMIN` / `VIEWER`).

SystemSettings

Singleton row (`Id = 1`) holding every admin-configurable setting across seven categories (General, Performance Review, Email, Authentication, Security, Dashboard, Branding). `SmtppasswordProtected` and `LoginAsVerificationPasswordProtected` are encrypted via ASP.NET Core Data Protection, never stored in plaintext.

EmailLog

One row per dispatch attempt. `Status` is one of `SENT` / `FAILED` / `LOGGED` (no SMTP configured)/ `DISABLED` / `SKIPPED_NO_RECIPIENT` / `SKIPPED_DUPLICATE` . `IdempotencyKey` prevents the same transition's email being sent twice.

AuditLog

Append-only administrative action trail. `EntityType` / `EntityId` let the Workflow Administration Details page pull exactly the entries for one review; `Action` carries a human-readable verb phrase (e.g. `"Workflow Administration: Return to Employee"`).

ImpersonationLog

One row per "Login As" session — opened on impersonation start, closed (`EndedAt`) on "Return to Administrator" or logout. Never deleted; this is the accountability record for that feature specifically, independent of (and not gated by) the general audit-logging toggle.

ScheduledJobRun

One row per execution of a Quartz job — necessary because the scheduler's own in-memory job store does not retain history across an application restart. Read by the Scheduled Jobs page for last-run status/duration/result.

Notification

One row per (user, event) — events affecting several people are fanned out into one row per recipient at creation time rather than modeled as a broadcast, keeping "mark as read" simple and per-user.

7. Indexes

TABLE	INDEX	PURPOSE
PmForms	(EmpCode, EvalYear) unique	One review per employee per year, enforced at the database level.
PmForms	LegacyRefNo unique	Legacy reference-number lookup.
PmForms	EvalYear	Dashboard/summary filtering by year alone.
PmForms	(EvalYear, Status)	The "this year, this status" pattern used by Workflow Administration search and PM Form Summary.
AuditLogs	(EntityType, EntityId)	Per-record audit history — hit on every Workflow Administration Details page view.
AuditLogs	OccurredAt , EmpCode , DeptCode , Action	Audit-trail filtering.
ManagerAssignments	ManagerEmpCode	Every "my team"/manager-filtered search and report.
AppUsers	UserName unique	Login lookup.
EmailLogs	IdempotencyKey , FormLegacyRefNo	Duplicate-send prevention; per-form email history.
ImpersonationLogs	SessionId unique, StartedAt	Session correlation; recent-history ordering.

8. Migrations

11 migrations, applied automatically on application startup (`Database.MigrateAsync()`), in order: initial schema, impersonation log, system settings, application base URL, missing-index pass, department description/active flag, Phase-9 settings and user security fields, audit log, scheduled job run, notifications, Phase-11/12 settings and preferred-culture, and the most recent missing-index pass (PmForm status/year composite, manager-assignment, audit entity-type/id).

APPLYING MANUALLY

```
dotnet ef database update --project src/PerformanceManagement.Core --startup-project
src/PerformanceManagement.Web
```

 — only needed if you deliberately disable the automatic startup migration.